From Tracking to Possession

A Reproducible Estimator for More Than A Hack 2027

Technical report for the Phase 1 submission

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Abstract

The task is to assign one of four labels to every frame of a football match: home possession, away possession, dead ball, or abstention. The central difficulty is not detecting the players; automatic tracking is supplied. It is inferring controlled possession from a noisy ball track, fragmented player tracks, and imperfect automatic tactical tags. This report explains a tracking-first estimator in physical and algorithmic terms. It combines pitch-metric proximity, temporal persistence, restart evidence, and explicit stoppage signals. The model is deterministic and reproducible, with fixed heuristic thresholds rather than learned parameters. It uses no video or human labels as runtime inputs.

1 What the challenge asks

For each frame t in the tactical recording, emit a label

$$y_t \in \{H, A, D, X\},$$

where H and A mean that the home or away team controls the live ball, D means that the ball is not in play, and X means that the estimator abstains. A possession is a maximal continuous run of one team's live-ball labels. A dead-ball interval terminates one possession and precedes another; it is not itself a possession.

The output is indexed by tactical-video frames and must contain exactly 199,319 rows, numbered 0–199,318. The match runs at 25 frames per second. Of these frames, 158,857 are graded. The supplied tracking is automatic and uncorrected; in particular, the ball is absent in about 41% of graded frames, and the 46 player track slots are temporary tracklets rather than persistent player identities.

1.1 Why the ranking metric changes the engineering problem

The ranking metric is *spellF1*. It compares whole possession intervals using temporal intersection-over-union and one-to-one matching, averaged over IoU thresholds 0.3–0.7. A system that predicts many short possession fragments can have reasonable frame labels and still miss the ranking objective. The first tie-break is macroF1, the unweighted mean of frame-level F1 for H, A, and D.

The scorer fills X with the previous non-X label within the same half before computing spellF1. Consequently, X is not a neutral gap at a transition: it can extend the old possession across a turnover. The estimator therefore carries its current team through ordinary tracking gaps, but does not use abstention as a substitute for deciding a detected change of control.

2 Available measurements and their interpretation

The model uses only the tactical source:

1. **FIFA/EPTS ATD metadata**: team membership of each track slot, frame rate, and pitch dimensions.
2. **Raw ATD**: normalized (x, y) positions of each tracked player and the ball, with optional speed measurements.
3. **Smart Tagging**: automatic phase intervals, team tags, set-piece events, and goal events. These tags are evidence, not ground truth labels.

The video is not decoded. The homography maps pitch coordinates to image pixels and adds no independent signal when no image pixels are used. The broadcast recording has a different, unsynchronized timeline and is omitted. These choices keep the pipeline lightweight and portable to another match with the same tactical data format.

3 The estimator as a noisy-measurement model

The conceptual state is the live possessor $s_t \in \{H, A\}$, with a separate dead-ball state. At each frame, the estimator asks two questions:

1. Is there evidence that play is stopped?
2. If play is live, which team has the stronger evidence of controlling the ball?

This is analogous to inferring a discrete physical state from noisy sensors. The ATD geometry is the primary sensor. Smart Tagging supplies a weak, correlated prior because it is generated automatically from the same match. Temporal persistence encodes the physical distinction between a touch and controlled possession. The numerical scores below are heuristic evidence scores, not calibrated probabilities.

3.1 Distance in pitch coordinates

The ATD positions are normalized to the pitch. Let the pitch length and width be L and W, the ball position be $(x_b(t), y_b(t))$, and a player position be $(x_i(t), y_i(t))$. Convert normalized coordinate differences to metres:

$$d_i(t) = \sqrt{[L(x_b(t) - x_i(t))]^2 + [W(y_b(t) - y_i(t))]^2}.$$

This anisotropic scaling matters: the pitch is rectangular, so an unscaled Euclidean distance in normalized coordinates would distort proximity, especially along the shorter axis.

For each team $g \in \{H, A\}$, let $d_g(t)$ be the smallest finite distance to a tracked player on that team. If the ball is not observed, the frame contributes no new team evidence. Otherwise,

$$e_g(t) = \begin{cases} \exp[-d_g(t)/\lambda], & d_g(t) < R, \\ 0, & d_g(t) \geq R, \end{cases} \qquad R = 4.5\,\text{m}, \quad \lambda = 1.8\,\text{m}.$$

The exponential gives a nearby player more weight than a distant one, while the finite range stops a far-away track from claiming the ball. The score is not a claim that the nearest player always controls the ball; it is an observation whose interpretation is stabilized in time.

3.2 Weak use of team phase tags

When an ordinary phase tag names a team, its score is increased by a small bonus $\beta = 0.25$ only if direct ball/player evidence already exists:

$$e'_g(t) = e_g(t) + \beta\, \mathbf{1}[e_g(t) > 0].$$

Overlapping tags do not accumulate without bound. A team tag by itself cannot invent possession when the ball is unobserved. This is important because the tagger is automatic and its phase intervals are not possession intervals.

4 Dead ball and restart logic

The estimator makes dead-ball decisions from event semantics and kinematics rather than treating every tagged phase as a stopped play.

4.1 Set-piece phase: stop first, then detect the restart

A `SET PIECES` interval can contain both the stoppage and live play after the restart. The algorithm therefore marks it dead only from the tag's start up to the first controlled-motion evidence. It uses the provider's ball speed where present. When speed is missing, it estimates speed from successive observed positions, preferring the longest available short lag (4, then 3, 2, or 1 frames):

$$\widehat{v}_b(t) = \frac{\|\mathbf{r}_b(t) - \mathbf{r}_b(t-k)\|}{k/\mathrm{fps}}.$$

A candidate restart requires all three of the following:

- the ball is observed;
- its measured or displacement-derived speed is at least $0.8\,\mathrm{m\,s^{-1}}$;
- the ball is within 4.5 m of a tracked player.

The first run of three consecutive candidate frames is taken as the restart boundary. This rejects single-frame motion noise and gives an operational meaning to "play has resumed." If no such run occurs inside the tagged interval, the interval remains dead-ball evidence through its end.

When a set-piece tag identifies the restarting team, it adds a weak bonus of 0.35 for 1.2 seconds after the detected restart, and only while direct ball/player evidence is present. For a `Kick Off (After Goal)` whose team is `N/A`, the restarting side is inferred as the opponent of the team in a nearby `GOALS` interval (within ten seconds of its end).

4.2 Goals and ball out of play

An automatic `GOALS` interval is treated as a dead-ball interval. Its time span runs from the scoring event through the stoppage and typically to the following kickoff. Independently, a ball track that lies beyond the normalized pitch rectangle by more than 0.005 for at least five consecutive frames is labeled dead ball over that sustained run. Shorter excursions are ignored to reduce sensitivity to coordinate noise. The latter rule is physically motivated by the ball having left the field, but it can confuse an airborne-ball localization error with a true out-of-play ball.

All dead-ball frames reset the live possessor. Evidence smoothing is done separately on the live segments on either side of a stoppage, so a previous team's evidence cannot leak through D.

5 Temporal decision rule for possession

Within each live segment, the team evidence is averaged over a centered radius of 0.28 seconds. Let the resulting scores be $\widetilde{e}_H(t)$ and $\widetilde{e}_A(t)$. A candidate is the team with the larger score. It must first have some support:

1. the larger score is at least 0.12.

If no team has yet been established, the first supported candidate initializes the state. Once a team is established, a change requires the opposing score to exceed the current team's score by at least 0.12 continuously for 1.2 seconds (30 frames at 25 fps). When confirmed, the algorithm backdates the change to the first frame of that uninterrupted evidence run. Brief touches or noisy nearest-player changes therefore do not cause a turnover, while a persistent recovery can.

If neither team has sufficient evidence, the current possessor is retained; before any possessor is established, the output is X.

In pseudocode, the live-state logic is:

```
if dead_ball:
    output D; clear current possessor
elif neither team has supported evidence:
    output current possessor, or X if none exists
elif no current possessor:
    initialize to strongest supported team
elif opposing advantage persists for 1.2 s:
    change state at the beginning of that evidence run
else:
    retain current possessor
```

The persistence interval operationalizes the challenge's notion of controlled possession. It is not a claim that all real possession changes take exactly 1.2 seconds; the interval is a generic noise filter intended to reject transient contact.

6 Implementation and reproducibility

The implementation is `possession.py`; it uses Python's standard library and NumPy. It reads team membership, frame rate, and pitch dimensions from the metadata, checks that ATD frame IDs start at one and are contiguous, and derives the output row count from the tracking file. No club name, match frame range, or team ID is hard-coded. The resulting CSV uses the exact header `frame,label` and enumerates frames from zero.

From the standalone package directory:

```
python3 -m pip install -r requirements.txt
python3 possession.py \
  --data-dir /path/to/match-data/tactical \
  --out submission.csv
python3 -m unittest -v test_possession.py
```

For this match, the generated `submission.csv` has 199,319 rows and passes the competition repository's submission validator. Re-running the estimator with the same inputs produced the same CSV byte-for-byte. The local CSV is ignored by the package's `.gitignore`; it should be uploaded separately as the possession-file deliverable rather than published in the code repository. The match data is not included in this package.

7 Validation history and compliance

The full-match private ground truth was not accessed and no human labels were used to train a model. However, public labels were used during development beyond checking file format and alignment: earlier versions were scored, their possession/dead-ball summaries were examined, the persistence duration was changed from 0.55 to 1.2 seconds after inspecting one result, and the public dead-ball prevalence informed later investigation of stoppage detection. This was use of the public labels to guide algorithm development, contrary to the repository's instruction that the public sample is for pipeline checking rather than tuning. The final revision has not been rescored, but it inherits that development history; it cannot be represented as having been developed independently of the public labels. There was no automated parameter sweep, but the manual changes still constitute public-label-informed tuning. Before submitting this result, the team should seek guidance from the organizers on whether the development history is acceptable.

The package includes 13 synthetic tests of invariants such as short touches not switching state, missing ball observations preserving the current possessor, restarts ending dead-ball states, and

smoothing not crossing a dead-ball boundary. These tests verify implementation behavior, not real-match accuracy.

8 Limitations and next evidence needed

The estimator's principal limitation is identifiability: proximity is not the same as control. A ball can pass close to an opponent without being intercepted, a rebound can resemble a controlled touch, and aerial ball coordinates can be unreliable. The 2D tracking omits some of the information that a human analyst uses, such as body orientation, contact, and whether a player deliberately played the ball. Smart Tagging is also generated automatically and shares some failure modes with the tracking.

Without labels from other matches, it is not possible to establish that the heuristic thresholds generalize optimally. A stronger scientific validation would use independently labeled matches (permitted by the rules), partition them by match, calibrate only on training matches, and report held-out spellF1 and macroF1 with uncertainty intervals. The public five-minute slice from this match is too small and is explicitly not a tuning set. Accordingly, this report describes a reproducible estimator and its assumptions, not a guarantee of winning or a claim of measured full-match superiority.

Competition references

The task definitions, allowed-use rules, file contract, and metric details are specified in the competition repository's `CHALLENGE.md`, `RULES.md`, `SUBMISSION.md`, `DATA.md`, and `EVALUATION.md`. The evaluator's metric implementation is the final authority for scoring.